

Axisymmetric Coanda-assisted vectoring

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Abstract An experimental demonstration of a jet vectoring technique used in our novel spray method called Coanda-assisted Spray Manipulation (CSM) is presented. CSM makes use of the Coanda effect on axisymmetric geometries through the interaction of two jets: a primary jet and a control jet. The primary jet has larger volume flow rate but generally a smaller momentum flux than the control jet. The primary jet flows through the center of a rounded collar. The control jet is parallel to the primary and is adjacent to the convex collar. The Reynolds number range for the primary jet at the exit plane was between 20,000 and 80,000. The flow was in the incompressible Mach number range ($Mach < 0.3$). The control jet attaches to the convex wall and vectors according to known Coanda effect principles, entraining and vectoring the primary jet, resulting in controllable $r - \theta$ directional spraying. Several annular control slots and collar radii were tested over a range of momentum flux ratios to determine the effects of these variables on the vectored jet angle and spreading. Two and Three-component Particle Image Velocimetry systems were used to determine the vectoring angle and the profile of the combined jet in each experiment. The experiments show that the control slot and expansion radius, along with the momentum ratios of the two jets predominantly affected the vectoring angle and profile of the combined jets.

1 Introduction

The present study on Coanda-assisted vectoring in axisymmetric geometries is motivated by the ongoing development of a spray manipulation technique called Coanda-assisted spray manipulation (CSM). The Coanda effect, or the tendency of jets to adhere to nearby surfaces (with a turning radius much larger than the jet size), is a well-established flow-control methodology. This traditional method is used in a blowing control flow to dominate a primary jet in order to provide profile and direction control and improve the stability of a jet, spray, or flame. As such, the new device will enable long-term operation of controllable jets or sprays in harsh, corrosive, or combusting environments, such as those associated with thermal sprays (Davis 2004). The key significance of this innovation is that the resultant device will make it possible to control the spray direction and several thermal-spray process parameters with a single nozzle and no moving parts in or near the flow (where combustion and/or high temperatures may be present).

The control jet is under the influence of the Coanda effect. The Coanda effect causes a jet to follow a curved surface if the radius of curvature of the surface is much larger than the jet (Newman 1961). This effect results from the reduced pressure on the inside of the turning radius. The reduced pressure effect competes with the dissipation of boundary-layer energy until the flow ultimately detaches from the surface. The blowing flow in CSM is applied approximately parallel to the primary flow and tangential to a curved collar, as shown in Fig. 1. The control flow attaches to the curved surface and influences the primary jet flow through momentum interactions. While the control jet by itself maintains a large vector angle, interaction with the primary jet causes the primary jet to vector as well.

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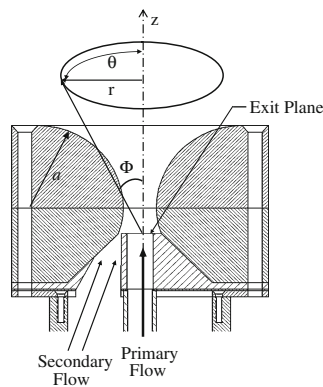


Fig. 1 Scale drawing of a Coanda-assisted vectoring nozzle. The application of control flow at one circumferential location will cause the primary jet to vector toward the control flow. By allowing the control location to be moved to an arbitrary θ location, and by varying the vectoring angle Φ , $r - \theta$ control over a spray flow can be achieved

This is shown below, as vectoring of the combined jet occurs for cases where the control flow momentum flux is large compared to that of the primary flow. A similar arrangement has been used on a planar geometry for thrust vectoring by Mason and Crowther (2002).

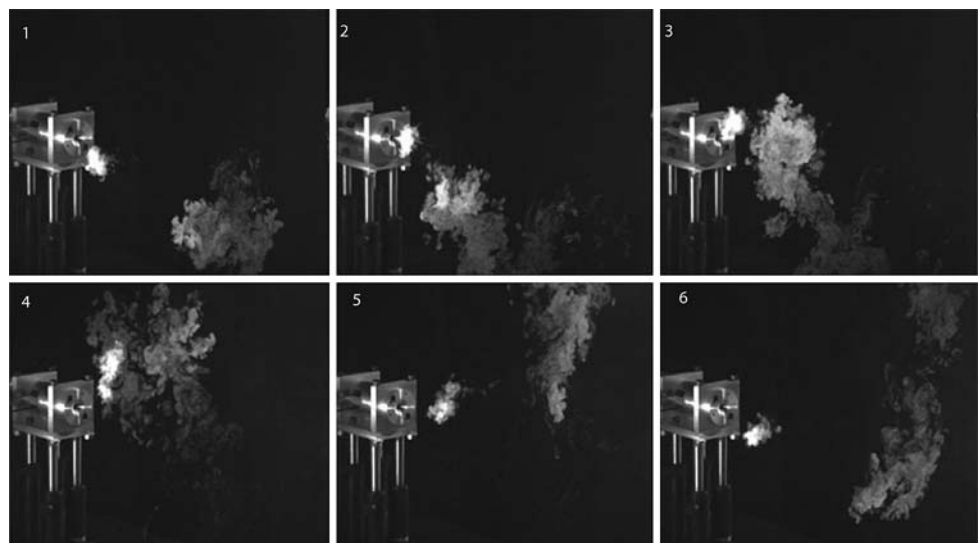
The important physics of the combined jet flow of the CSM technique has two parts. The first is the Coanda effect of the control jet interacting with the primary jet in planar geometry as described above. The second is the interaction of the combined jets with the three-dimensional surface of the convex collar. The jet is no longer acting in two-dimensional space, so shear-layer instabilities will occur as the turbulent control jet attaches to the surface and entrains the turbulent primary jet. These instabilities are modified when the geometry of the exit slot is changed, as shown in the results below, with the vector angles of the 15.1% circumference slot test behaving quite differently than other slot sizes.

With the Coanda-assisted vectoring scheme applied to a jet flow, the vectored jet can be rotated rapidly by modifying the circumferential position at which the control flow is applied. Using CSM, a jet or spray's angle can be altered constantly to maintain an orthogonal relationship to the coated surface. Coatings sprayed orthogonal to the surface have been found to exhibit higher microhardness, higher compressive residual stress, and less surface wear than off-angle spraying methods (Strock et al. 2001). The magnitude of the vectoring and the profile of the main jet are controlled through varying the momentum flux ratio between the control jet and the primary jet. The nozzle rotates to provide rotational direction, θ in Fig. 1, and is the only moving part on the device.

Other fluidic jet vectoring schemes may not require Coanda surfaces, but typically require larger control flows and combinations of blowing and suction such as demonstrated by Smith and Glezer (2002), Bettridge et al. (2006), Humes and Smith (2008) and Hammond and Redekopp (1997). Vectoring using a Coanda surface and a synthetic jet control was demonstrated by Pack and Seifert (2001). However, synthetic jets are complicated and unreliable. Strykowski et al. (1996) vectored a high-speed jet using an extended surface and control slot through which air was sucked. The ingested air may be at high temperature and may cause structural failure. Therefore the method of CSM was chosen as a simple, highly controllable, stable, maintainable solution.

A CSM concept device was demonstrated at the Experimental Fluid Dynamics Laboratory (EFDL) at Utah State University. Flow visualization movies of this device are posted on the EFDL website (Smith 2007). Several frames from the demonstration are shown in Fig. 2. The jet was seeded with oil droplets that were illuminated by two laser sheets at different downstream locations. The control

Fig. 2 Six frames of the demonstration video for CSM. The applied control flow is ramping up while being rotated in the clockwise direction. The full video can be found at <http://www.efdl.usu.edu>



flow was ramped up from zero while rotating the nozzle at a fixed rate. This resulted in the jet starting in the center and then spiraling with an increasing radius. The device was found to work over a broad range of speeds with the results depending primarily on the ratio of the primary to control jet momentum.

The demonstration unit, scaled similarly to our vision of a production CSM unit, had a primary jet exit diameter of 3.2 mm. The jet exited at the center of the collar and the collar radius to primary jet diameter was $a/D = 3.00$. The secondary slot extent was 37% of the circumference. The secondary slot width was 0.9 mm, the gap between the two jets was 0.2 mm, and center of the collar was 4.8 mm in diameter. The maximum vector angle achieved was about $\Phi = 50^\circ$.

The Coanda effect has been widely used in fluidics (Joyce 1983) with both aeronautics and medical applications, air moving technology and other fields. Nevertheless, this phenomenon is not completely understood, especially for three-dimensional flow as in the CSM concept design. The nature of the Coanda effect, with boundary layer separation and entrainment interaction, make it difficult to solve the flow analytically. In fact, (Wille and Fernholz 1965) claimed that there was no unique solution to this type of flow. Therefore, most work on the subject is based on experiments [although numerical studies have been performed and are gaining momentum as computational methods improve (see Sawada and Asami 1997; Wernz and Fasel 2006)]. Rask (1973) and Patankar and Sridhar (1972) have studied two-dimensional flows around cylindrical surfaces, looking at flow characteristics in all three dimensions (normal to surface, laterally across curved surface, and streamwise). However, to our knowledge, no research has been performed on surfaces other than two-dimensional geometries.

In order to determine the geometric and flow parameters affecting CSM control, the Coanda effect in axisymmetric geometries must be first understood. The present experiments investigate the variation of vectoring angle and jet spreading for a non-rotating axisymmetric Coanda-assisted flow as a function of the exit geometry and flow parameters and will provide guidance toward developing more effective CSM. The experimental facility and measurements will be described followed by results and conclusions.

2 Experimental facility

A sketch of the test setup is shown in Fig. 3. The primary (A) and the control (B) high-pressure-air flows entered separately at the rear of the test setup. The primary flow was channeled via a conduit (E) directly through the setup to the collar (H) at the exit. The control flow entered a

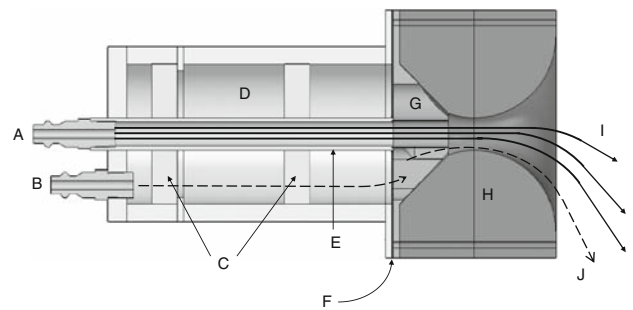


Fig. 3 Sketch of experimental test CSM facility. The *solid streamlines* represent the primary flow, while the control flow is indicated with *dashed lines*

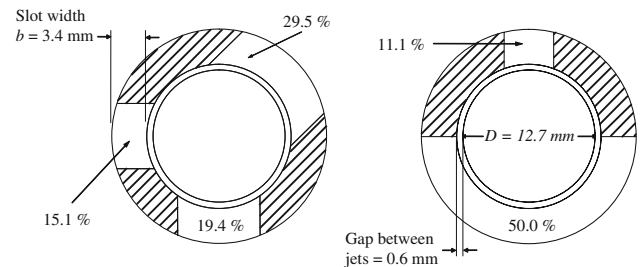


Fig. 4 Sketch of the control slots as viewed looking into the jet exit. All control slots had the same width, but different circumferential extents, shown as percentages of the total circumference. One slot was used for each test

plenum (D) that contained two flow conditioning sections (C) in order to spread the flow evenly. From the plenum, the flow was directed to a control slot (G), and a gasket (F) was used to close off slots that were not being used (each part G had several different slots, only one of which was used at any time). The exiting control flow was vectored (J) around the collar (H) by the Coanda effect and through entrainment, the primary flow (I) was vectored as well.

The test facility was designed to allow several different geometric variations, including the collar radius a , and the circumferential extent of the control slot. Figure 4 shows the geometries of the slots at the exit plane graphically from a view looking into the jet. Two different parts comprising a total of five different control slots were used.

The primary jet, which had a diameter of $D = 12.7$ mm, emerged from a circular exit. The control slots were annular, with the slot width $b = 3.4$ mm. Circumferential percentages (based on the smaller radius) of 11.1, 15.1, 19.4, 29.5, and 50.0% were used. The gap between the jets at the exit plane was constant at 0.6 mm. The collar had a diameter of 15.9 mm at the most narrow point. The primary and control jets exited at a point one diameter upstream of the center of the collar (see Fig. 3, notice the location of the jet exits is upstream of the most narrow point of the collar).

Two, independent, Porter model 204A mass flow controllers were used in the experiment. The primary mass flow controller was rated 0–1000 SLPM, while the control jet mass flow controller was rated 0–200 SLPM; each with uncertainty of $\pm 1.5\%$ full scale. The primary flow air was infused with olive oil particles using a seeder and plenum while the control flow was pure air. The momentum rates were controlled independently, and many values of momentum flux ratio were tested to view their effect on vectoring angle. The form of the momentum flux ratio, assuming uniform flow at the exit, is

$$J^* = \frac{J_c}{J_p} = \frac{\dot{m}_c^2}{A_c \rho} / \frac{\dot{m}_p^2}{A_p \rho}, \quad (1)$$

where J_c is the control momentum flux, J_p is the primary momentum flux, A_c is the control exit slot area, A_p is the primary exit slot area, $\dot{m}_p$ is the primary mass flow rate, $\dot{m}_c$ is the control mass flow rate, and ρ is the fluid density. Since the Mach numbers of the jets are low, the densities of the two jets at the exits are assumed equal.

3 Measurements

Two separate Particle Image Velocimetry (PIV) systems were employed in the testing. The first was a single camera two-component system. The 12-bit, 1.3-MPixel ($1,280 \times 1,040$) camera acquired images of seeded flow illuminated by a pair of 532 nm wavelength, 50 mJ Nd:YAG lasers with sheet optics. The second PIV system was a stereo (two camera) system that returns three-dimensional velocity components. The laser was the same used in the two-component setup and the two cameras were identical to that used for the two-component measurements. Both systems used LaVision DaVis software to compute the velocity vectors. The final interrogation regions were 16×16 pixels. The time between images was set to ensure that the maximum particle displacement was near 4 pixels.

For the two-component measurements, the beam was cast perpendicular to the CSM exit plane. The control flow was applied at a location such that the jet always vectored in a direction inside the laser plane (see Fig. 5a). These measurements were used primarily to determine the vector angle. For the stereo measurements, the beam was directed parallel to the CSM exit plane at a fixed distance downstream (see Fig. 5b). These measurements were used primarily for determining the resultant mixing of the jet.

Table 1 shows the matrix of geometries tested. The radius of the collar, a , was varied between 25.4 and 66.7 mm and the control jet varied from 11 to 50% of the total circumference. These values were chosen for easy machining. For each control jet exit slot geometry, the

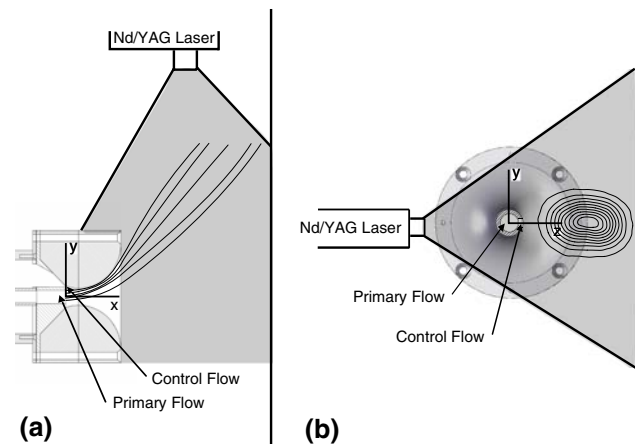


Fig. 5 Laser sheet configuration of two-dimensional (a) and stereo (b) systems

control mass flow rate to primary mass flow rate was varied between 0.0 and 1.0. In doing so, the momentum flux ratios varied between 0.00 and 7.75. The flow was in the incompressible Mach number range ($\text{Mach} < 0.3$). The state of the exiting flow from each jet was assessed with hot-film measurements. A probe was placed first in the center of the primary jet at its exit, and then in the center of the control slot. Over the full range of conditions tested, the streamwise fluctuation levels in both jets varied from 4% to 14% of their means. The fluctuations in the primary jet were unaffected by the control jet, while the control jet fluctuations increased when the primary jet was flowing. Additionally, spectra of the velocity signals were examined to ensure that the flow was turbulent. Based on these measurements, both jets are turbulent at their exits in all cases reported here.

For each combination of geometry and jet momentum flux, 100 two-dimensional PIV image pairs were taken and averaged to generate mean velocity vector fields. A sample averaged velocity vector field is shown in Fig. 6.

When determining the vector angle of the jet, we chose not to include the entrainment regions of the jet. For each averaged velocity vector field, the standard deviation of velocity magnitude was calculated and used as the threshold velocity

Table 1 Geometric variation matrix (2D = test via two-component PIV, 3D = test via three-component PIV)

a/D	2.00	2.75	3.25	5.25
Slot size				
11.1%	2D	2D	2D, 3D	2D
15.1%	2D	2D, 3D	2D	2D
19.4%	2D, 3D	2D	2D	2D, 3D
29.5%	2D, 3D	2D	2D	2D, 3D
50.0%	2D	2D, 3D	2D, 3D	2D

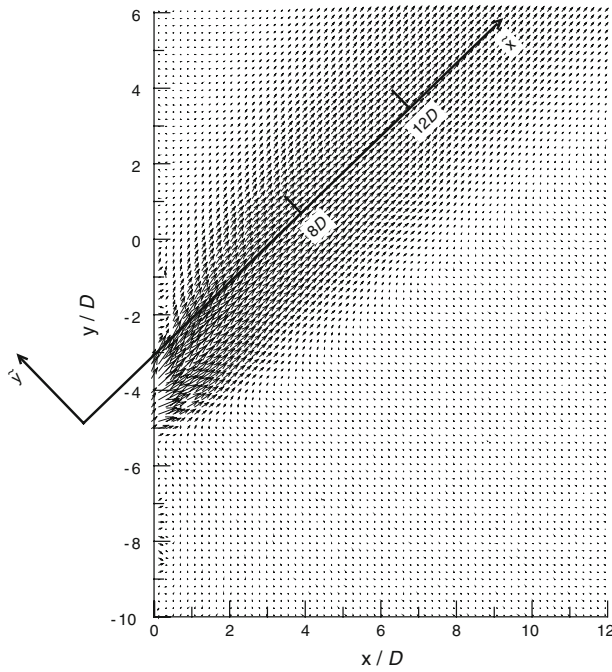


Fig. 6 Velocity vector field for circumferential percentage of 29.5%, $a/D = 2.00$, and $J^* = 0.769$. The coordinate system $(\tilde{x}, \tilde{y})$ is also shown

to eliminate the entrainment regions of the mean velocity field. The average vector angle of the jet was calculated by averaging the angle for all locations where the mean jet velocity magnitude exceeds the threshold velocity. The coordinate system was then shifted so that the x coordinate was parallel to the jet vectoring angle. This coordinate system is denoted $(\tilde{x}, \tilde{y})$ and has corresponding velocity components $\tilde{u}$ and $\tilde{v}$. The chosen origin of rotation was the center of the primary jet exit. Note that the collar extends downstream of this point, however this initial angle is used only to generate a plane along which the momentum components were calculated. At $8D$ downstream of the jet exit along this line, the $\tilde{x}$ and $\tilde{y}$ momentum components were calculated according to $\int \rho \tilde{u}^2 d\tilde{y}$ and $\int \rho \tilde{v}^2 d\tilde{y}$, respectively. The resultant vector angle is deemed to be the vector angle of the jet. Thus, an initial angle guess is made, then improved by the momentum vector. The vector angle for the case in Fig. 6 was found to be $\Phi = 49^\circ$.

4 Uncertainty analysis

The sensitivity of this angle calculation to the threshold velocity was determined by choosing three different threshold velocities (standard deviation, and standard deviation plus or minus 10% of the standard deviation). The resultant vector angles for these choices were all

within 4° and the standard deviation of difference in angle was only 0.52° .

The uncertainties in velocity as acquired by the two-component PIV are dominated by precision errors rather than bias errors since only 100 image pairs were acquired. The uncertainty in the magnitude of velocity with 95% confidence is found according to

$$u_{|V|} = 1.96 \frac{S_{|V|}}{\sqrt{100}}, \quad (2)$$

where $u_{|V|}$ is the uncertainty in the velocity magnitude and $S_{|V|}$ is the standard deviation of the magnitude of velocity. Scaling this uncertainty by the half maximum velocity finds that the largest uncertainties in velocity magnitude are 2–8% of the measured velocity.

The vectored jet area, width, height, and growth rate were examined using stereo PIV data. Data was collected at $\tilde{x} = 8D$ and at $\tilde{x} = 12D$ for three-component tests listed in Table 1. For each geometric setup and jet momentum flux ratio, 300 PIV image pairs were acquired simultaneously by the two cameras at 5 Hz. LaVision DaVis software was used to generate averaged three-component velocity vector fields from these images. A FORTRAN code was used to detect the cross-sectional width and height of the jet profile at half the maximum velocity based on the averaged velocity vector fields. It was observed that the jet cross-sectional profile was elliptical, so the cross-sectional areas of the jets were found by fitting the equation for an ellipse using the measured width and height. The uncertainties of the velocity in the z -direction of the three-component PIV data are found using the same method as the two-component PIV data and are found to range between 4 and 9% of the measured velocity w .

The uncertainty in the momentum flux ratio arises from the uncertainties in the exit slot areas due to machining errors and the uncertainty of the mass flow controllers used throughout the experiments. Since density cancels in Eq. 1, the uncertainty in the momentum flux ratio is found according to

$$U_{J^*}^2 = U_{A_c}^2 \left(\frac{\partial J^*}{\partial A_c} \right)^2 + U_{A_p}^2 \left(\frac{\partial J^*}{\partial A_p} \right)^2 + U_{\dot{m}_c}^2 \left(\frac{\partial J^*}{\partial \dot{m}_c} \right)^2 + U_{\dot{m}_p}^2 \left(\frac{\partial J^*}{\partial \dot{m}_p} \right)^2. \quad (3)$$

Calculating the partial derivatives in Eq. 3 and simplifying, the uncertainty equation becomes

$$U_{J^*} = J^* \sqrt{\left(\frac{U_{A_c}}{A_c} \right)^2 + \left(\frac{U_{A_p}}{A_p} \right)^2 + 4 \left(\frac{U_{\dot{m}_c}}{\dot{m}_c} \right)^2 + 4 \left(\frac{U_{\dot{m}_p}}{\dot{m}_p} \right)^2}. \quad (4)$$

The area of the primary slot was found according to $A_p = \pi D^2/4$. The area of the control slot was found according to

$$A_c = l * w + \frac{r_{\max}^2}{2}(\alpha_{\max} - \sin \alpha_{\max}) - \frac{r_{\min}^2}{2}(\alpha_{\min} - \sin \alpha_{\min}), \quad (5)$$

where l is the length of the slot, w is the width of the slot, $r_{\max}$ is the larger radius, $\alpha_{\max}$ is the central angle of the larger radius in radians, $r_{\min}$ is the smaller radius, and $\alpha_{\min}$ is the central angle of the smaller radius in radians.

Since the control slot area was varied, the uncertainties in the control slot areas range from 3 to 5%, while the primary slot area has an uncertainty of 0.02%. At most, the area uncertainties account for less than 10% of the total uncertainty. The majority of uncertainty comes from the primary and control mass flow rates which are 1.5% full scale. As the accuracy was based on full scale percentages, the uncertainty for those flow ratios on the lower end of the operating range are substantial and plots of the data show error bars to indicate this uncertainty.

5 Results

The vectoring performance in terms of vector angle as a function of the momentum flux ratio for various collar radii and control slot sizes is shown in Fig. 7. The vectoring angle of the primary jet is a strong function of the non-dimensional momentum flux ratio. The angle is zero with no control flow and generally increases with momentum ratio to a maximum before either decreasing slightly or becoming asymptotic. Mason and Crowther (2002), in their planar geometry experiment similar to this, found a similar trend. Mason and Crowther also experienced the “reverse Coanda effect” found at very low momentum ratios in which the primary jet actually vectors toward the opposite curved wall. This is shown in the figure by the negative angles present at very small momentum ratio values. As stated by Mason and Crowther, in the “reverse Coanda effect” regime, “The faster moving primary jet entrains the slower moving air of the secondary jet and instead of adhering to the Coanda surface and remaining attached far downstream, the secondary jet separates and a high pressure region forms. The entrainment has the effect of skewing the primary jet velocity distribution towards the opposite Coanda surface and the differential pressure gradient created vectors the primary jet in that direction.”

Also shown in the plots is the effect of the radius of the collar. The vector angle increases proportional to the radius. The effect is substantial for the smaller control slots, while the effect of radius is small in the largest control slot. Insensitivity to the collar radius will be advantageous in situations where one desires to vector the flow along a radius that will allow dense particulate to remain inside the

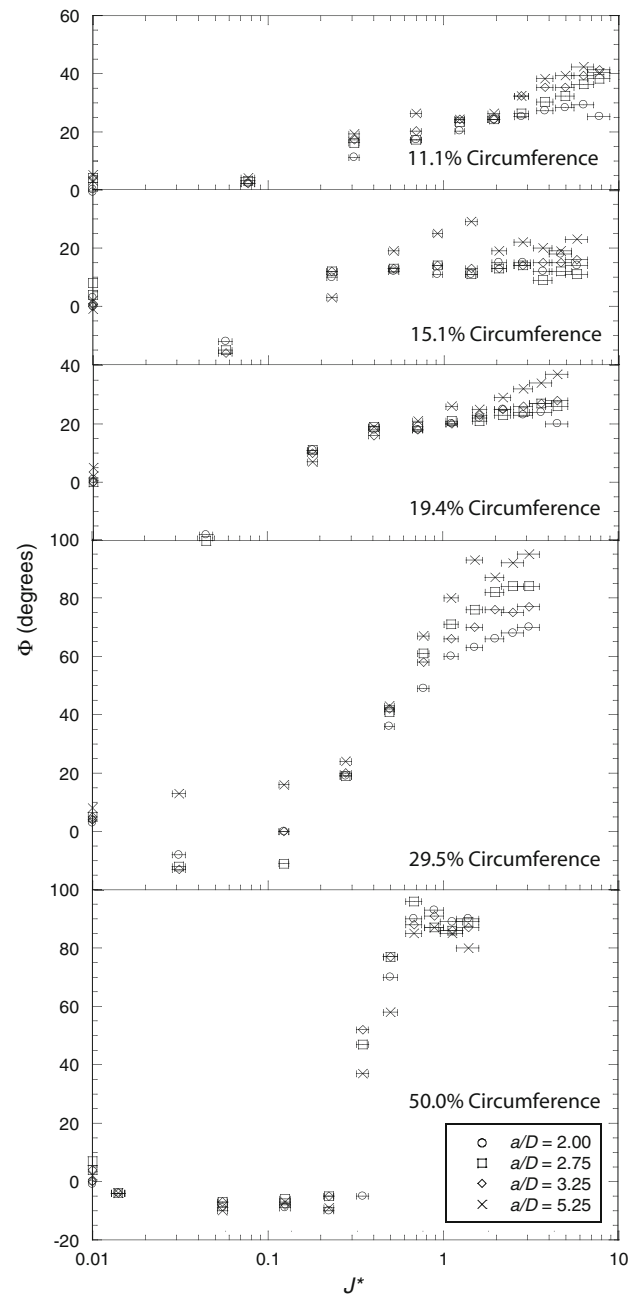


Fig. 7 Vector angle as a function of momentum flux ratio for several values of collar radius and control slot size

vectored flow. The area of the control slot also strongly influences the peak and minimum angles.

For each slot size, beyond the flux ratios that resulted in negative angles, the angle increases to a peak value. This behavior is generally linear and the initial slope of these curves is larger for the larger control slots. This can be seen more clearly by examining the momentum ratio necessary to achieve 90% of the maximum angle for each control slot. As shown in Fig. 8, the momentum ratios corresponding to the maximum angles become smaller with the

size of the control slot. Although Mason and Crowther (2002) did not have an analogous parameter to the control slot circumference to compare to the slope of the linear region, they did mention a similar “control region” where the vectoring may be effectively controlled according to momentum flux ratio. This plot also shows the large sensitivity to collar radius for small control slots and the insensitivity of the vectoring to collar radius for larger control slots. The data is not completely monotonic as the 15.1% circumference slot does not follow the trend. This is assumed to be the effect of triggered or suppressed instabilities that may be altering the profiles and thus changing the stability of the jets in a non-axisymmetric way.

A correlation function was sought which would enable the accurate prediction of vector angle for given momentum ratios, collar radius, and exit slot circumference percentage. Newman (1961) described the form of a function that would predict the separation angle (angle at which the jet detaches from the curved surface) of a single jet over a Coanda surface to be

$$\Phi_{\text{sep}} = f \left[\left\{ \frac{(p - p_{\infty})ba}{\rho v^2} \right\}^{1/2} \right], \quad (6)$$

with $p - p_{\infty}$ being pressure differential (where p is the supply pressure and p_{∞} is the free stream pressure), b is the exit slot width, and a is the radius. Using this angle function as a guide, our correlation function incorporates momentum ratio rather than pressure difference as the driving force. Based on the results in Figs. 7 and 8, the

results are scaled on the control slot size and the best agreement is found using

$$\Phi = \left(C1 \times J^* \times \frac{a}{D} \right)^{\left(\frac{\lambda}{C2} \right)}, \quad (7)$$

where λ is the control slot arc length divided by the primary slot circumference. The constants, $C1$ and $C2$, have values of 3.25 and 2.0, respectively. The results of this correlation function are shown in Fig. 9, with various geometries specified.

We note that the different control slots may have very different impacts on the jet and its stability, which may explain some of the data that does not fit the correlation well. With 50% of the jet surrounded by the control slot, the results are expected to (and do) resemble those for a 2-D geometry (Mason and Crowther 2002). For very small control slot such as 11.1% of the circumference, less control authority is obtained, and the scaling of the correlation accounts for this. The two slot geometries with somewhat dissimilar behavior are those with slots that are 15.1 and 19.4% of the circumference. We conjecture that these cases lie between two regimes. It is possible that the modification of the boundary layer of the primary jet fluid for these parts of the circumference leads to enhancement of circumferential instabilities that inhibit the vectoring effect. Unfortunately, our data set is not sufficient to confirm this conjecture.

With the relationship between the geometry, momentum ratio, and the resultant vectoring angle established, we turn our attention to the impact that this vectoring scheme has on the mixing of the jet. The velocity profile and half-

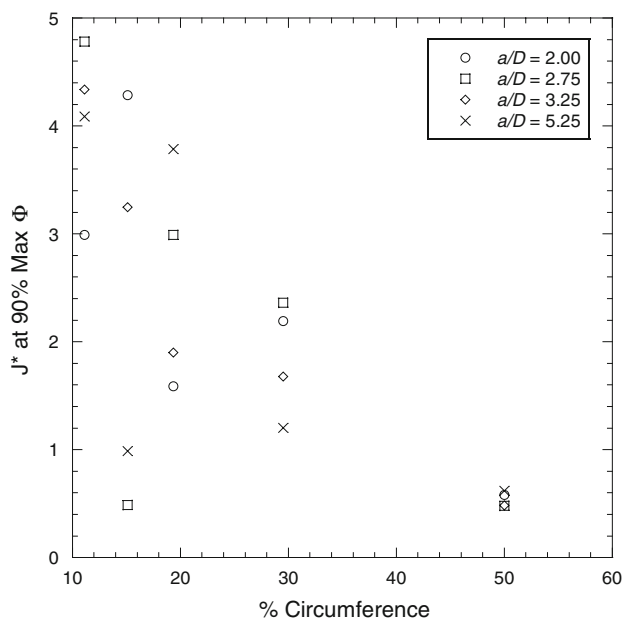


Fig. 8 Momentum ratio at 90% of maximum angle as a function of slot size

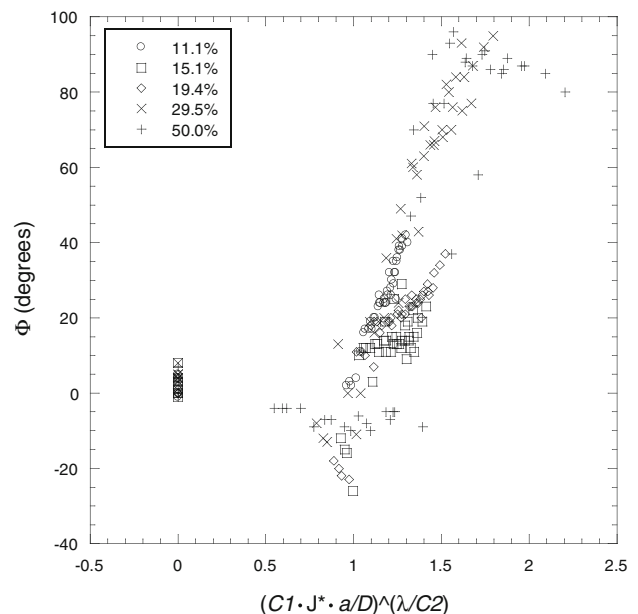


Fig. 9 Vector angle as a function of correlation formula including all test data, outlined according to slot size

maximum width of the vectored jet at $\tilde{x} = 8D$ downstream of the primary jet exit were examined using two-dimensional PIV. Two baseline measurements were taken for each geometry, one with zero control blowing and primary mass flow rate of 0.004 kg/s ($Re = 20,400$) and another with zero control blowing and primary mass flow rate of 0.011 kg/s ($Re = 61,300$). These baseline profiles were compared with vectored jet profiles having the same primary mass flow rate. The average baseline vector angle is 2.63° , apparently due to small variations in the machining.

To examine the jet spreading, and thus the mixing, the maximum half-widths were calculated for each of the experiments and the percent difference of the vectored to unvectored maximum half-widths was calculated. The average percent difference for the lower velocity jets was found to be 19.6%, while the higher velocity jets were more similar with an average percent difference of 9.5%. The profile shape changes with increasing momentum flux ratio are shown in Fig. 10. The maximum half-width (indicated by the dashed lines), normalized by D , ranges from 2.25 for no blowing to 4.32 for the largest flow ratio ($J_c = J_p = 3.08$). Also, the maximum velocity location shifts from near the centerline to below the centerline or away from the curved surface adjacent to the control slot. At the same time, the shape also becomes asymmetric with the maximum velocity tending toward the Coanda (positive y) surface. It is noted that the velocity profiles are not complete across the $\tilde{y}$ axis, especially for profiles in the highly vectored cases. This is due to the rotating of the x axis according to vector angle and plotting the velocity profile perpendicular to that axis.

The cross-sectional area of the jet at $8D$ and $12D$ downstream of the exit plane was measured using stereo PIV. Contour plots of one geometric combination and J^* are shown in Fig. 11. The control flow was applied to the top of the primary jet in this figure. For this case, the area of the vectored jet is 4.07 times greater than the unvectored jet area.

The areas (based on half-maximum velocity) at $8D$ and $12D$ of vectored jets are compared with unvectored jets in Fig. 12. Note that there are no data for the larger vector angles for the 29.5% circumference and $a/D = 5.25$ at $8D$ downstream since part of the jet left the measurement domain. However, the behavior at the two locations is similar. We find that for all but the 29.5% circumference and $a/D = 5.25$ case, the cross-sectional area increases slowly with vector angle until the saturation angle is reached, and then increases at a higher rate. Most importantly, the area of the jet for a 29.5% circumference and $a/D = 5.25$ is not significantly impacted by the control until very large vector angles, demonstrating that this vectoring scheme can be achieved without significantly mixing the jet.

Although the normalized areas varied for the two downstream locations, the aspect ratio was not affected by distance downstream as illustrated in Fig. 13 where both $8D$ and $12D$ data are plotted. The data is decreasing, or, in other words, the height increases more rapidly than the width, for angles up to 90° . In terms of the CSM geometry, the jet profile is stretching in the $\tilde{y}$ -positive direction or towards the control slot. After this point the width increases extremely fast with increasing angle. In other words, the jet spreads more rapidly across the Coanda surface as vector angle is increased past $\Phi = 90^\circ$. This is most likely due to the vectored jet interacting with the exit plane and changing its shape as a result.

6 Conclusions

A fluidic jet-vectoring scheme termed “Coanda-assisted Vectoring” was demonstrated on an axisymmetric geometry. For the parameter space including $20,000 < Re < 80,000$ and $Mach < 0.3$, the vectoring performance of 28 geometric combinations was examined. The control/primary momentum flux ratio was varied in the range from 0.00 to 7.75. The

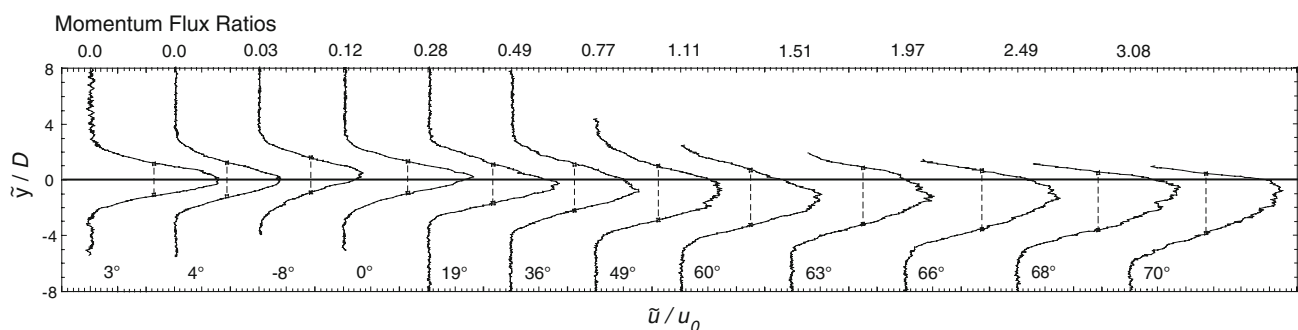


Fig. 10 Velocity profiles at $\tilde{x} = 8D$ with increasing momentum flux ratio ($J^* = 0.0$ through 3.08) for 29.50% circumference and $a/D = 2.00$. The maximum half-width and vectored angle for each momentum ratio are also shown

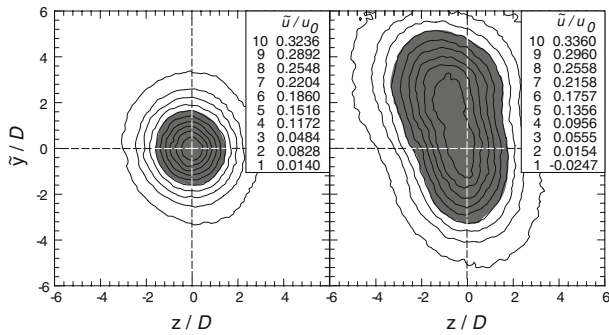


Fig. 11 Contour plots of the streamwise velocity non-dimensionalized by the exit velocity in a plane normal to the jet at $\bar{x} = 12D$ for 29.5% circumference and $a/D = 2.00$. The left plot is no vectoring (zero control blowing), while the right plot is with vectoring ($J^* = 2.49$, $\Phi = 70^\circ$). The velocity contours above half the maximum velocity are shaded

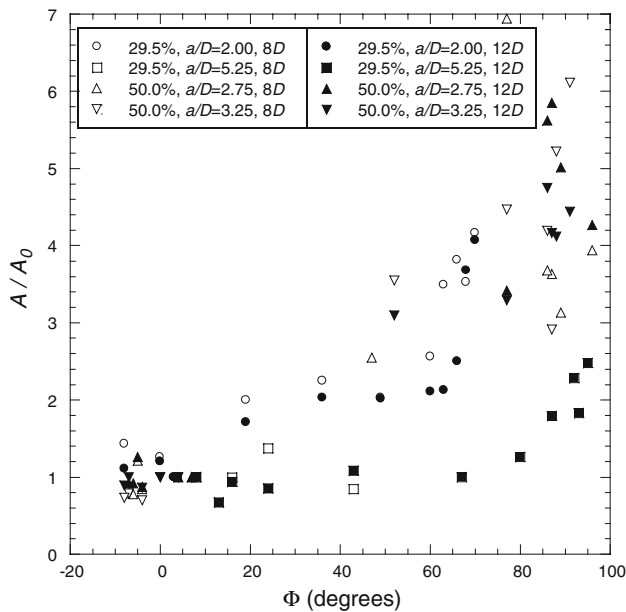


Fig. 12 The area of the jet normalized by unvectored jet area at $\bar{x} = 8D$ and $\bar{x} = 12D$ as a function of vector angle for various geometries

maximum vector angle was found to be 96° . The fundamental behavior of the vectoring angle with momentum flux ratio was similar for most values tested: for very small momentum ratios, negative vectoring was observed; after vectoring in the positive direction (toward the control flow), the vectoring angle increased with momentum ratio to a peak value; then angle either decreased slightly or increased at a much smaller slope with increasing momentum ratio. The control jet exit slot area was found to have a strong effect on the maximum angle and the momentum ratios required to achieve the maximum vectoring angle. The slope of the linear portion of the vectoring angle as a function of momentum ratio increased with slot size. The vectoring

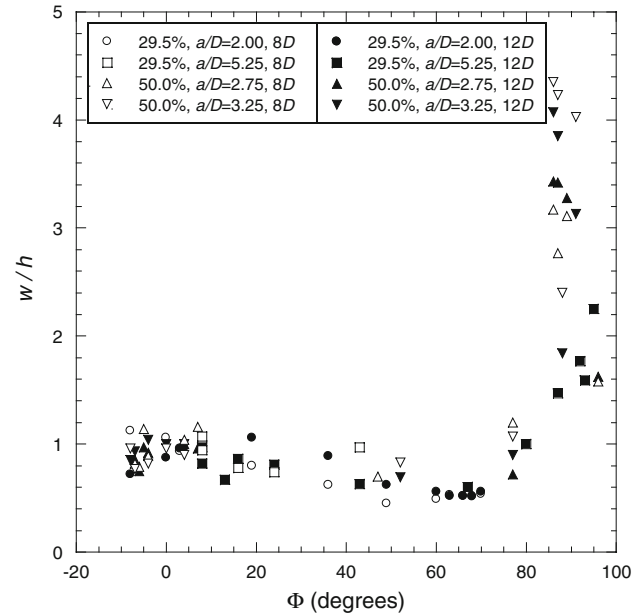


Fig. 13 Aspect ratio (jet width w divided by height h) of the vectored jets at both $\bar{x} = 8D$ and $\bar{x} = 12D$ for various geometries as a function of vector angle

angle was found to increase with collar radius, with a greater increase for smaller slots. A correlation function, scaled on the control slot size and following a separation angle equation presented by Newman (1961), was shown to describe the smallest and largest control slots well.

The velocity profiles for vectored and unvectored jets were similar in shape for low momentum flux ratio cases. With increasing momentum ratio, vectoring caused the maximum velocity to shift away from the Coanda surface and the velocity profile to widen. The profile became increasingly asymmetric with momentum flux ratio.

Vectored flow cross-sectional area increased proportional to vector angle at $8D$, while exceeding the saturation angle caused rapid growth in area at $12D$ downstream of the primary jet exit. The aspect ratio remains constant until $\Phi = 90^\circ$, after which the width of the jet increases dramatically.

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